



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

GITLAB CI/CD PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A DevOps

TOTAL: 10 QUESTIONS

Q1 What problem does GitLab CI/CD solve in DevOps or infrastructure?

Threat Model

Q6 How does GitLab CI/CD integrate with CI/CD pipelines?

Observability

Q2 What is GitLab CI/CD's core architecture?

Architecture

Q7 How do you debug a failed GitLab CI/CD deployment?

Lifecycle

Q3 How does GitLab CI/CD define infrastructure or configuration as code?

Configuration

Q8 What are security best practices for GitLab CI/CD?

Compliance

Q4 How does GitLab CI/CD ensure idempotency?

Hardening

Q9 How does GitLab CI/CD scale for large environments?

Testing

Q5 How do you handle secrets in GitLab CI/CD?

SSO / IdP

Q10 What are common mistakes teams make when adopting GitLab CI/CD?

Tuning



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 GitLab CI/CD addresses a specific concern

Mitigation

GitLab CI/CD addresses a specific concern - provisioning, configuration management, orchestration, or CI/CD - by automating manual steps that are error-prone, slow, or not reproducible when done by hand.

A6 CI/CD pipelines call GitLab CI/CD in

Observability

CI/CD pipelines call GitLab CI/CD in automated steps: plan on a pull request to preview changes and apply on merge to main. Approval gates can require a human to review the plan before changes go to production.

A2 GitLab CI/CD has a control plane

Primitives

GitLab CI/CD has a control plane that stores desired state and a data plane (agents or push-based SSH) that enforces it. Understanding this distinction clarifies where configuration is evaluated and where changes are applied.

A7 Check GitLab CI/CD's logs for the

Automation

Check GitLab CI/CD's logs for the specific error message and the resource that failed. For infrastructure tools, re-run with increased verbosity (-v, --debug). Review the state file or event history to understand what changed before the failure.

A3 GitLab CI/CD uses declarative files (YAML,

Hardening

HCL, or a DSL) to express desired state. A plan/diff phase shows what will change before applying. Version-controlling these files enables peer review and rollback.

A8 Rotate credentials used by GitLab CI/CD,

Governance

Rotate credentials used by GitLab CI/CD, restrict network access to its control plane, enable audit logging of all operations, and use least-privilege service accounts. Never run GitLab CI/CD with admin credentials in production.

A4 GitLab CI/CD operations are designed to

Gotchas

GitLab CI/CD operations are designed to produce the same result when run multiple times. Applying the same config twice converges to the desired state rather than creating duplicate resources. This makes automation safe to re-run.

A9 Large GitLab CI/CD deployments use

Assessment

Large GitLab CI/CD deployments use workspaces or namespaces to isolate environments, remote state backends for team collaboration, and modular code to avoid a single monolithic config that is slow to plan/apply.

A5 Secrets should never be stored in

Federation

Secrets should never be stored in plain-text in GitLab CI/CD config files. Use a secrets backend (Vault, AWS Secrets Manager) and inject values at runtime via environment variables or encrypted vaults supported by the tool.

A10 Common pitfalls include storing secrets in

Optimization

Common pitfalls include storing secrets in plain-text config, skipping the plan step before apply, not reviewing state drift, and not having a rollback strategy when a deployment fails partway through.